
Beyond Behavioral Fit: Tracing the Internal Mind's Trajectory via Modular Bayesian Inference in Human Category Learning

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Abstract

Modeling internal information processing from observable behaviors is essential to understanding human cognition. Yet such inference remains challenging, in part due to the subjectivity of the human mind: similar behavioral outputs may stem from distinct internal processes across individuals. To address this, we propose a modular Bayesian framework that performs trial-resolved inference while capturing two major sources of individual variability: bounded rationality (via perceptual noise and memory decay) and personalized prior biases. We apply our framework to a demanding context: tracing each individual's learning trajectory in a four-dimensional feature space for novel category learning, governed by three categorization rules of increasing complexity. We found, through comprehensive ablation studies, that although several reduced variants closely approximate the participants' choice accuracy, only the full model faithfully reconstructs the trial-by-trial evolution of each learner's internal exploration within the feature space. This underscores the risk of using external behavioral fit alone as a proxy for cognitive validity. Parameter analyses further reveal an adaptive analysis-by-synthesis mechanism: agents dynamically adjust their prior sampling within the feature space to balance exploration and exploitation, compensating for inherent memory limitations and enabling robust learning from limited data. Our work contributes both methodological advances to individualized cognitive modeling and algorithmic insights into flexible learning under bounded rationality, with implications for designing human-inspired AI systems and for tracking fine-grained internal processing dynamics across cognitive tasks.

1 Introduction

Understanding the internal representations of the human mind is central to uncovering the mechanisms of cognition, from what sensory information is selectively attended to construct memory [29], to what specific information is predicted to guide adaptive decision-making [6, 2, 27]. Although existing models of prediction, sensory coding, memory, and decision making are implicitly framed within the information processing paradigm [16], precisely tracking how this processing unfolds within the 'black box' of the mind remains a considerable challenge.

This challenge stems from the inherently subjective nature of cognition [1], which introduces both variability and ambiguity into the modeling process. First, individuals exhibit subjective experiences and idiosyncratic biases that shape how information is prioritized [30], resulting in considerable variability in internal cognitive processes [24]. Second, this variability creates an ill-posed mapping between high-dimensional sensory inputs and low-dimensional behavioral outputs (e.g., identity judgments such as 'Tom'). Multiple, mechanistically distinct algorithms can yield indistinguishable

behavioral responses (e.g., one observer may rely on nasal prominence and chin morphology, while another integrates nasal features with skin tone). Third, this ill-posed nature challenges conventional modeling approaches that fit canonical algorithms to population-averaged behaviors. Inferring algorithmic solutions *a priori* and evaluating models solely based on their ability to simulate average behavior risks overlooking optimal, individualized solutions, thereby hindering the discovery of novel cognitive phenomena and the advancement of mechanistic theory.

Contributions. We propose a data-driven framework that explicitly models trial-resolved signatures of internal information processing at the individual level. Our main contributions are: (1) **A modular inference framework** that approximates internal information processing as Bayesian inference, with separate modules for cognitive constraints that impose bounded rationality and personalized priors reflecting heterogeneous experience [3, 9]. (2) **Trial-resolved inference** in a demanding context—novel category induction over a high-dimensional feature space—demonstrating the model’s capacity to recover subject-specific feature exploration over the course of learning. Crucially, our model operates over a theoretically unconstrained hypothesis space and produces high-fidelity, trial-wise estimates of internal processing dynamics. (3) **Epistemological insights into computational modeling** via ablation studies, which reveal a key dissociation—models can reproduce observable behavior to a high degree while mischaracterizing internal representations. This highlights a fundamental epistemological risk of using behavioral fit alone as a proxy for cognitive validity. (4) **Mechanistic insights into human learning behavior** through computational dissection, which identify an adaptive analysis-by-synthesis strategy: agents dynamically tune their prior-sampling distributions to balance exploratory hypothesis search with exploitative evidence gathering, offsetting memory capacity and enabling robust learning from sparse data.

Related works. Categorization sits at the intersection of cognitive science and computer science: behavioral experiments examine how task structures shape human category representations [22, 17, 19, 11], while machine learning emphasizes maximizing classification accuracy through large-scale image-classification benchmarks such as ImageNet and deep architectures from AlexNet to ResNet [5, 20, 10, 8]. To uncover the dynamics of category learning, recent work adopts three main complementary modeling frameworks: (1) **Reinforcement-learning (RL) models** treat learning as prediction-error–driven weight updates on an explicitly defined value function, capturing shifts in human rule weighting but relying on strong assumptions about states, values and update rules [25, 3, 4]. (2) **Bayesian models** frame categorization as probabilistic inference over structured hypothesis spaces: hierarchical non-parametric priors (e.g., Chinese-restaurant or Dirichlet processes) compress the rule space, enabling one-shot generalization, latent-feature inference, and trial-by-trial predictions of feature-specific attention [21, 26]. (3) Shallow **neural-network models** such as ALCOVE and SUSTAIN view category learning as an information processing stream—from feature input layer through attention weighting to response output layer—where attention weights progressively focus on diagnostic features [15, 14]. However, all these works reconstruct internal trajectories only retrospectively from behavior correlations, raising the question of whether such post-hoc reconstructions truly capture each individual’s internal states. This highlights the need for frameworks that can record and explain within-subject learning dynamics in real time [7, 13].

2 Human experiments of category learning

The review board of our institution provided ethical approval for all the human experiments listed below. All participants received monetary compensation above the local minimum wage for their participation.

2.1 Methods

We designed a categorization learning task that encourages participants to explore a four-dimensional feature space of novel visual stimuli. Each stimulus was a line drawing, parametrized by four variable-length segments with fixed angles (Fig. 1A). Participants were trained to perform correct categorization following three different rules of increasing complexity (Fig. 1B, between-subject design): (1) a two-category rule based on a single randomly assigned feature (e.g., leg); (2) a four-category rule requiring integration of three features (e.g., leg, tail, and head), with the hierarchical structure disclosed; and (3) a structurally similar four-category rule, but without revealing the underlying hierarchy (see Appendix A for rule details). Illustrated using the two-category task, each

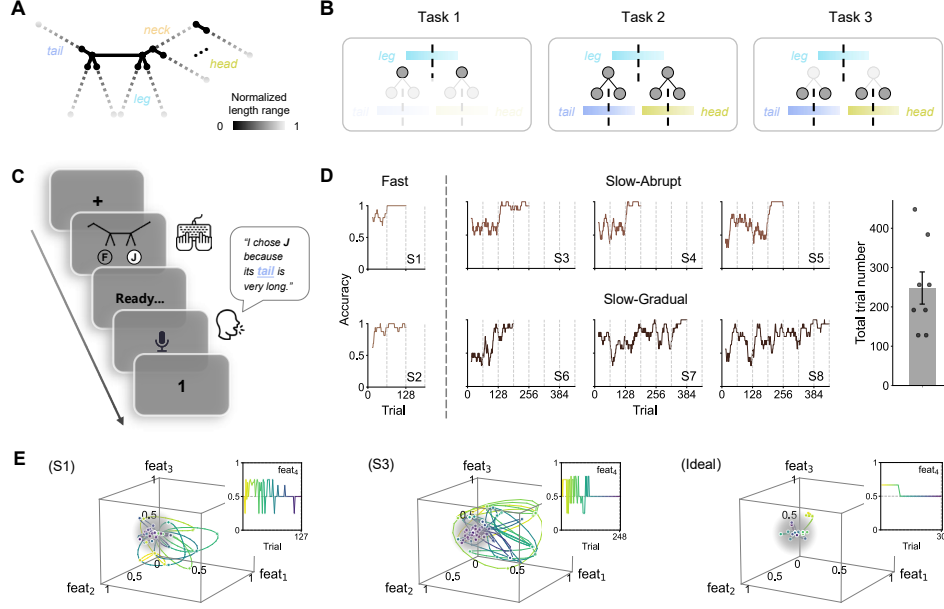


Figure 1: Stimuli, task, and human behaviors. **(A)** Each stimulus has four independent length features normalized to the range 0–1. **(B)** Between-subjects tasks increase in complexity: Task 1 (two categories, single feature); Task 2 (four categories, three-feature hierarchy shown); Task 3 (same rule as Task 2, hierarchy hidden). **(C)** On each trial, participants observed a randomly sampled stimulus, selected a category (e.g., F or J in Task 1), verbally reported the feature(s) that informed their choice (indicated by a speech bubble), and received deterministic feedback. **(D)** Learning performance in Task 1 showed large individual differences: participants varied widely in the total number of trials needed to reach criterion and in sliding-window accuracy (line plots), which could be broadly classified into three groups—fast (lightest brown), slower but abrupt (medium brown), and slower and gradual (darkest brown). **(E)** Exploration trajectories in feature space from two representative participants (S1, S3) trained on the same task rule illustrate early broad sampling followed by late local refinement (as shown by the yellow-to-purple gradient), whereas the model-derived ideal observer (right) converges rapidly.

trial (Task 1 in Fig. 1C) required participants to classify a randomly generated stimulus into one of two arbitrary categories (“F” or “J”), following a deterministic rule unknown to them. After each response, participants verbally reported the features they used to make the choice, serving as a ground truth for internal processing. Feedback was then provided (1 = correct, 0 = incorrect), followed by the next trial. Learning concluded once participants achieved over 90% accuracy across 64 consecutive trials. Before learning, all participants completed a calibration task to quantify their baseline perceptual noise (see Appendix A).

2.2 Results

Participants generally succeeded in all three tasks, with three excluded due to persistent failures. However, the number of trials needed for successful learning varied considerably across individuals within the same task (Fig. 1D, boxplot). More notably, learning dynamics—quantified using sliding-window accuracy—also differed markedly: even in the simplest two-category task, participants ranged from gradual learners (lighter hues) to those exhibiting abrupt transitions (darker hues) (Fig. 1D). Similar variability was observed in other more complex tasks (see Appendix A).

To probe internal processing, we analyzed participants’ verbal reports projected into the four-dimensional stimulus space (see Appendix A for methodological details). Feature-based trajectories from two exemplar participants trained on the same rule are shown in Fig. 1E (left and middle), from earlier (yellowish) to late (darkish) trials. This visualization reveals two key observations: (1) an early phase of broad, saltatory exploration, followed by a later phase of localized refined

108 exploration; (2) substantial individual variability in these dynamics, which diverge markedly from an
 109 ideal observer (Fig. 1E, right)—implemented via the baseline model described in Section 3.1—who
 110 integrates evidence optimally and converges rapidly on the correct rule.

111 **Summary.** Our findings reveal idiosyncratic learning patterns across participants, evident in both
 112 observable performance (Fig. 1D) and internal hypothesis trajectories (Fig. 1E). A detailed analysis
 113 of feature exploration suggests a strategic process—participants appear to engage in active inference
 114 via simplified prior assumptions and iterative hypothesis testing—reflected in the shift from saltatory to
 115 localized sampling. Notably, these trajectories often diverged from the optimal path, underscoring the
 116 bounded rationality of human information processing. Together, these results highlight the necessity
 117 of modeling internal dynamics to uncover the underlying mechanisms of human learning.

118 3 The modular Bayesian framework

119 We develop a computational model to infer each learner’s internal processing states—specifically, their
 120 use of stimulus features during category learning—based solely on behavioral responses. Notably, the
 121 verbal reports collected during learning serve exclusively for model validation, providing **ground-**
 122 **truth benchmarks** to evaluate the model’s performance in predicting humans’ internal states. As
 123 illustrated in Fig. 2, we formulate the learner’s internal processing as trial-resolved Bayesian inference
 124 (Fig. 2B) over a rich hypothesis space (Fig. 2A), enabling comprehensive and coherent utilization of
 125 both stimulus properties and behavioral variability. We also implements three cognitively motivated
 126 components (perception, memory, and hypothesis generator) into the inference process to capture
 127 the bounded rationality of human learners (Fig. 2C). This modular Bayesian framework generates
 128 eight model variants through module assembly (see Fig. 3A), enabling systematic analysis of
 129 how individual cognitive factors—both independently and interactively—shape the learner’s inference
 130 trajectory and observed behaviors, while enhancing explanatory power for the empirical phenomena.

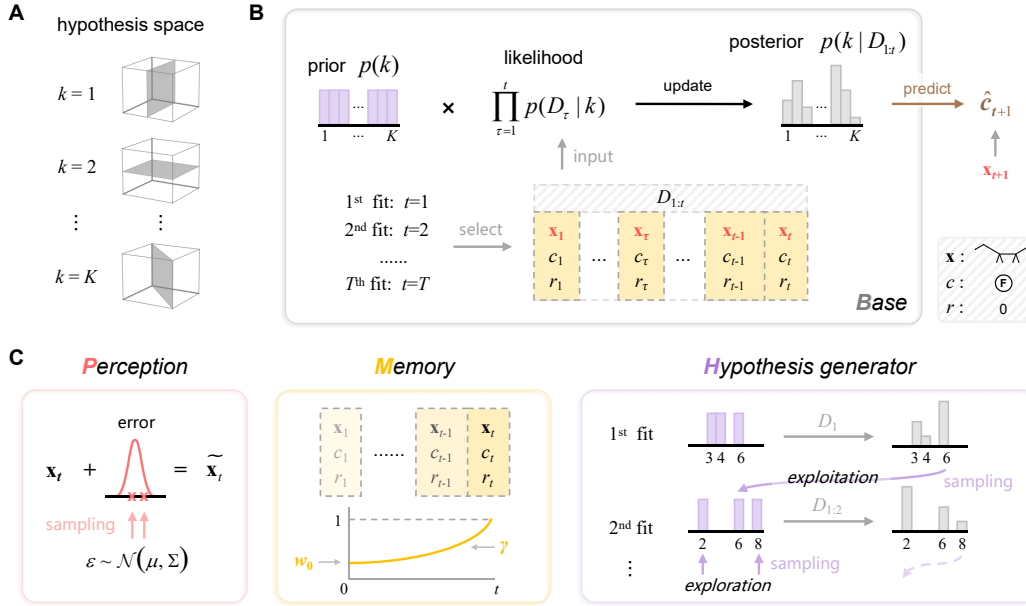


Figure 2: Schematic of the Modular Bayesian framework. **(A)** Hypothesis space illustrated in a 3D feature space with two categories. Each hypothesis divides the space into two regions and assigns category probabilities to stimuli. **(B)** Bayesian inference engine in the rational backbone, serving as the baseline. Agents update their beliefs over hypotheses by integrating stimulus, choice, and feedback data. The resulting posterior is then used to predict the choice on the next trial. **(C)** Three bounded-rational modules augment the core inference in (B): a perception module adds Gaussian noise to stimulus input, a memory module down-weights past evidence via exponential decay, and a hypothesis generator adaptively samples a working hypothesis set through exploration–exploitation trade-offs.

3.1 Rational backbone

We posit that each learner entertains a library of geometric “partition” hypotheses that carve the four-dimensional feature space into regions associated with category labels, following classic concept-learning works showing that humans preferentially generate and test explicit rules (Bruner, Goodnow & Austin 1956; Feldman 2000) rather than tuning infinitesimal weights. More importantly, Bayesian belief updating over this hypothesis space provides a normative engine that converts stimuli into graded category probabilities and refines those probabilities from trial to trial.

Hypothesis space. The *hypothesis space* $\mathcal{H}$ is defined as a family of soft *Voronoi* partitions mapping the *feature space* $\mathbf{X} = [0, 1]^D$ (here $D = 4$) to the *category probability space* Δ^{N-1} (with $N = 2$ in Task 1 and $N = 4$ in Tasks 2&3). Each parametric partition $\theta \in \mathcal{H}$ is parameterized by a unique hard *Voronoi* partition k , serving as the prototype, and a sharpness-coefficient $\beta > 0$ that controls the softness of the decision boundary, thus $\theta = (k, \beta)$. For simplicity, we further restrict k to a finite set $\mathcal{K}$ of geometrically regular partitions (see Fig. 2A for an illustration in 3D space and Appendix B) and treat β as an optimizable parameter, so that $|\mathcal{H}| = |\mathcal{K}|$. This keeps the hypothesis space expressive enough to approximate arbitrary decision surfaces, yet finite enough for efficient Bayesian updating on a trial-by-trial basis.

Bayesian update. Agents are initialized with a uniform prior over the discrete hypothesis prototypes, $p(k) = 1/|\mathcal{K}|$, and a uniform prior over the sharpness parameter, $\beta \in [\beta_{\min}, \beta_{\max}]$, which is used exclusively for optimization. By Bayes’ rule, the posterior over hypotheses θ after observing trial t is updated as $p(\theta | \mathbf{x}_t, c_t, r_t) \propto p(\mathbf{x}_t, c_t, r_t | \theta) p(\theta)$, where $p(\theta)$ denotes the current prior belief over hypothesis θ and $p(\mathbf{x}_t, c_t, r_t | \theta)$ is the likelihood of the observed stimulus, choice, and feedback on trial t . This likelihood can be factorized as:

$$p(\mathbf{x}_t, c_t, r_t | \theta) = p(r | \mathbf{x}_t, c_t, \theta) p(c_t | \mathbf{x}_t, \theta) p(\mathbf{x}_t | \theta). \quad (1)$$

Here, $p(c_t | \mathbf{x}_t, \theta)$ denotes the probability that a learner holding hypothesis θ would produce the observed choice c_t when presented with stimulus $\mathbf{x}_t$. This probability is computed via the distances between the stimulus and the category centers specified by the partition k . $p(r_t | c_t, \mathbf{x}_t, \theta)$ and $p(\mathbf{x}_t | \theta)$ are defined by our experimental setup and independent of θ . See Appendix B for further details.

Aggregating evidence up to trial t (included), denoted as $\mathcal{D}_{1:t} = \{(\mathbf{x}_\tau, c_\tau, r_\tau)\}_{\tau=1}^t$, the log-posterior over hypothesis θ can be computed as (see Fig. 2B):

$$\log p(\theta | \mathcal{D}_{1:t}) = \log p(\theta) + \max_{\beta} \left(\sum_{\tau=1}^t w_{\tau} \log p(D_{\tau} | \theta) \right) - \log Z_t, \quad (2)$$

where $Z_t = p(\mathcal{D}_{1:t})$ is a normalization constant. The likelihood is w_{τ} -weighted to account for imperfect memory (see Section 3.2): $w_{\tau} = 1$ corresponds to fully rational Bayesian updating with perfect memory, while $w_{\tau} \leq 1$ mimics human forgetting by down-weighting past evidence.

3.2 Bounded-rational adaptations

Three auxiliary modules are included to reconcile optimal computations with realistic human constraints (see Fig. 2C). Specifically, the Perception module (P-module) accounts for uncertainty in sensory coding; a Memory module (M-module) captures information decay over time; and a Hypothesis generator module (H-module) imposes inductive bias by restricting the generation of candidate hypotheses.

Perception. We assume that the features perceived by participants are corrupted by an independent Gaussian noise channel,

$$\tilde{\mathbf{x}}_t = \mathbf{x}_t + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad (3)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ denote the feature-wise perceptual bias and variability, respectively, for each individual. These parameters are estimated from the Perception Calibration task (see Section 2 and Appendix A).

Memory. To account for imperfect memory, we down-weight older evidence via an exponential decay function, rescaling the log-likelihood sum in Eq. (2):

$$w_j = w_0 + \gamma^{t-j}, \quad 0 \leq \gamma, w_0 \leq 1. \quad (4)$$

Here w_j quantifies the contribution of trial j ($1 \leq j \leq t$) to the current log-posterior update at trial t . The parameter γ controls the decay rate, with smaller values indicating faster forgetting, while w_0 captures a baseline memory trace—analogueous to a long-term store that resists decay [12]. By fitting (γ, w_0) separately for each participant, the model can express a spectrum of recency effects and capture systematic individual differences in memory fidelity.

Hypothesis generator. Given cognitive capacity limits, learners likely operate on a dynamically updated subset of hypotheses—rather than maintaining the full hypothesis space as assumed in the rational backbone—as suggested by explorative behavioral patterns in Section 2. We implement this process based on pre-defined sampling strategies and dynamically adjusted hypothesis-set sizes.

We define three strategies for hypothesis selection: *posterior-based exploitation* (S_1), *cue-driven associative recall* (S_2), and *stochastic exploration* (S_3). At every trial t the learner firstly decides the amount of hypotheses to draw from each strategy, $(m_1^{(t)}, m_2^{(t)}, m_3^{(t)})$, then generates the next working hypothesis set $\mathcal{H}_{t+1} = \bigcup_{i=1}^3 S_i^{(t)}$ from

$$\begin{aligned} S_1^{(t)} &\in \left\{ (\text{Top-}m_1^{(t)})(q_t(k); \mathcal{H}_t), \quad \text{No-Rep}(\pi(q_t(k)); m_1^{(t)}) \right\} \\ S_2^{(t)} &\in \left\{ (\text{Top-}m_2^{(t)})(s_t(k|\mathbf{x}_{t+1}); \mathcal{H}_t \setminus S_1^{(t)}), \quad \text{No-Rep}(\pi(s_t(k|\mathbf{x}_{t+1})); m_2^{(t)}) \right\} \\ S_3^{(t)} &\sim \text{Uniform}(\mathcal{K} \setminus (S_1^{(t)} \cup S_2^{(t)}), m_3^{(t)}) \end{aligned} \quad (5)$$

where $q_t(k)$ is the current posterior and $s_t(k)$ denotes the associative similarity between the next stimulus $\mathbf{x}_{t+1}$ and partition k (see details in Appendix B).

Remarkably, due to constraints inherent in human behavioral experiments, learning is typically observed as a one-shot trajectory sampling process—it is infeasible to collect repeated data from the same condition. Following prior work [7], we incorporate a repeated sampling mechanism into the perception module and hypothesis generator to mitigate the randomness introduced by single-sample trajectories.

3.3 Model fitting and comparison

Model space. The full model space comprises the rational backbone (serving as the baseline model) combined with one, two, or all three bounded-rational adaptations, yielding eight candidate models. Systematic ablations allow us to attribute predictive gains to specific cognitive constraints.

Optimization algorithms. We illustrate the optimization procedure using the full model (PMH-model). For each participant, we conduct a grid search over memory parameters (γ, w_0) to maximize the posterior likelihood of their observed behavior. The resulting best-fit parameters are then used to generate 1000 simulated behavioral trajectories, from which the one that best matches the participant’s behavioral profile is selected (see Appendix B for details). For other reduced model variants, the same procedure is applied, with the corresponding cognitive module(s) omitted during optimization.

Algorithm 1 PH_0M_0 Model

Input: $\{(\mathbf{x}_t, c_t, r_t)\}_{t=1}^T, \mathcal{H}, \gamma, w_0$
Output: $(\hat{c}_t)_{t=1}^T, \{p_{post}^{(T)}(k)\}_{t=1}^T$
Initialize: $p_{post}^{(0)}(k) = \frac{1}{|\mathcal{H}|}$
1: **for** $t \leftarrow 1$ to T_{max} **do**
2: $\hat{x} \leftarrow x + \epsilon$
3: $p_{post}^{(t+1)}(k) \leftarrow \text{Bayesian}(p_{post}^{(t)}(k), \mathcal{H}, \gamma, w_0)$
4: $(\hat{c})_t \sim \mathbb{E}_{p_{post}^{(t+1)}(k)}(\text{choice}(c \mid h, \vec{x}))$
5: $\mathcal{H} \leftarrow \text{Generate } \mathcal{H}_{new}$
6: **end for**

Predictive assessment. We fit models using only behavioral data, while verbal reports collected during learning were used solely for validation, providing ground-truth benchmarks of participants’ internal states. To evaluate model alignment with both external behavior and internal representations,

we computed two metrics: the mean absolute difference in learning performance (measured via sliding-window accuracy), and the difference in the probability assigned to the true hypothesis—estimated from model posteriors and participants’ verbal reports, respectively. Lower values on both metrics indicate better model-human alignment (see Appendix B for additional model fit indices).

4 Model Experiments and Results

4.1 Capture learning trajectories via model fitting and ablation

Our modular Bayesian framework can simulate a wide range of learning dynamics, allowing us to isolate latent factors that shape individuals’ learning behavior. Fig. 3A–B summarizes model performance across eight candidate variants using the two metrics introduced in Section 3.3. While the full model (HPM) shows comparable external performance to several reduced models (M, PM, MH, and PMH), it outperforms them markedly in capturing internal representations (** indicates FDR-corrected $p < 0.01$).

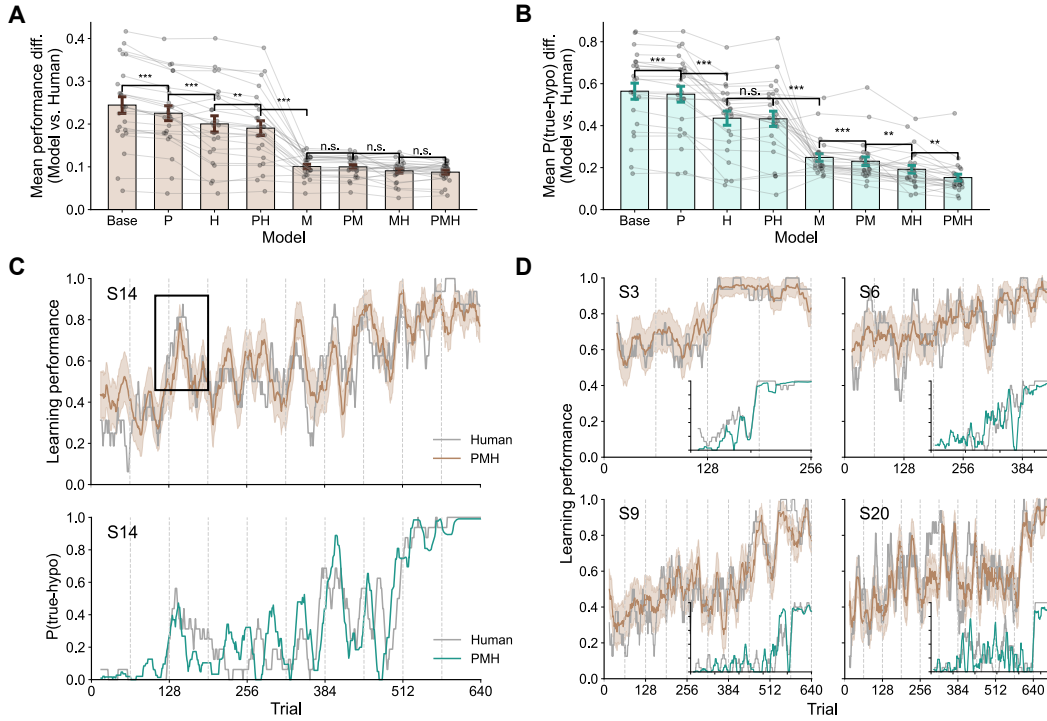


Figure 3: Model performance in predicting external behavior and internal representations. (A–B) Model comparison across eight candidate variants. Lower values indicate better model-human alignment. (C) Trial-by-trial fit for an exemplar participant using the HPM-model. Upper panel: Model-predicted learning performance (brown) closely matches the participant’s actual accuracy (gray). Lower panel: internal hypothesis trajectory predicted by the HPM-model (blue) aligns well with the ground-truth trajectory derived from the participant’s verbal reports (gray). (D) Generalization of the HPM-model across individuals and tasks. The model robustly recovers diverse internal trajectories, capturing individual-level variability in learning strategies.

We focus on three variants (i.e., HPM-model, M-model, and H-model) whose behavior offers particularly informative insights (see Appendix B for results for the remaining models). We first fitted each participant’s learning session using the full model (HPM-model). The upper panel of Fig. 3C shows the fit for an exemplar participant S14, comparing the model-predicted learning performance with the participant’s actual performance as a measure of alignment in overt behavior. As shown, the HPM-model (brown) closely follows the participant’s learning trajectory (gray), capturing not only the overall trend but also fine-grained local fluctuations with high fidelity (see highlighted region).

228 The performance fluctuations likely reflect ongoing hypothesis switching over stimulus features,
 229 leading to oscillatory patterns in the learning performance. To evaluate this interpretation, we
 230 computed the internal hypothesis trajectory (i.e., the utilization of features in the space) as the hit
 231 accuracy (y-axis) of agent’s active hypothesis (e.g., long leg) with respect to the target rule (e.g., short
 232 neck) using a sliding window over trials (x-axis). The lower panel of Fig. 3C shows the trajectory
 233 as predicted by the HPM-model (blue), which closely tracks the ground-truth trajectory obtained
 234 from participants’ verbal reports (gray). This strong alignment provides compelling evidence for
 235 the model’s validity in capturing internal representations, particularly notable given that no model
 236 parameters were fitted to the verbal data. Fig. 3D further demonstrates the HPM-model’s robustness
 237 across individuals, accurately recovering diverse internal trajectories despite substantial inter-subject
 238 variability. Taken together, these results show that the HPM-model successfully accounts for both
 239 the external behavior (i.e., category choices) and internal hypothesis dynamics (i.e., evolving feature
 240 representations), thus capturing the individual idiosyncrasies in the learning process.

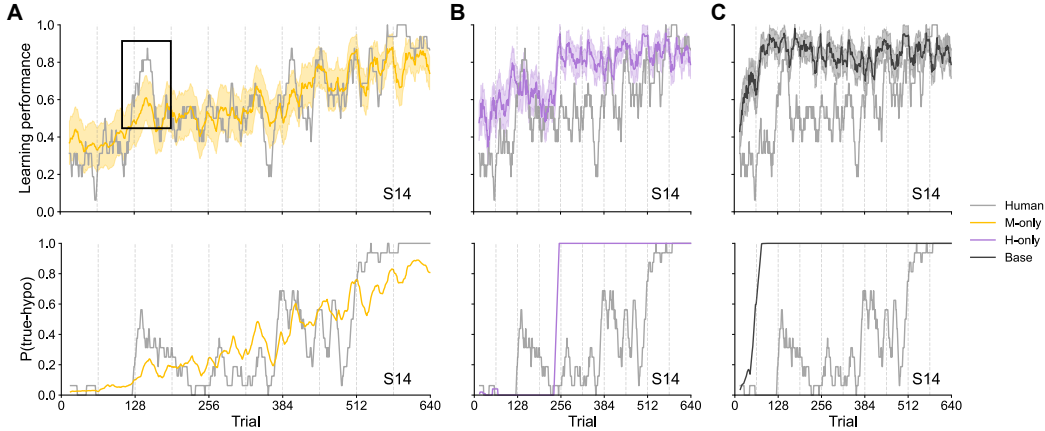


Figure 4: Behavioral and representational fits for two reduced model variants. **(A)** M-only model generally captures human-level behavioral accuracy but fails to align with participants’ internal hypothesis trajectories. **(B)** H-only model fails to reproduce both behavioral performance and internal dynamics, resembling the overly rapid convergence of the base model. **(C)** Base model’s trajectory illustrates near-optimal rule acquisition without bounded constraints.

241 Fig. 4A-B presents critical comparisons using two representative models from our ensemble: the
 242 M-model, which implements only the memory decay mechanisms, and the H-model, which includes
 243 only the hypothesis generator. These reduced variants were chosen to highlight two interesting cases.
 244 For the M-model, it predicts the overall trend of human overt behavior (see yellow vs. gray line) at a
 245 similar level to the HPM-model (Fig. 3C). However, its prediction on the internal hypothesis trajectory
 246 diverges significantly from participants’ verbal reports (see small panel in Fig. 3D). This dissociation
 247 highlights a critical point: modeling overt behavior alone proves insufficient – even good-level
 248 performance predictions may fundamentally misrepresent the internal processing. In contrast, the
 249 H-model (Fig. 4B) fails to reproduce both behavioral performance and internal hypothesis dynamics
 250 of the human learner. Its learning trajectory resembles that of the base model (Fig. 4C), which rapidly
 251 converges to the categorization rule. These results suggest that the local hypothesis sampling and
 252 rapid switching-and-testing are unnecessary for an agent with perfect memory and sensory fidelity.
 253 However, for humans, frequently switching among feature hypotheses in high-dimensional spaces
 254 may serve as a compensatory strategy to mitigate information loss (due to perceptual noise and
 255 memory decay), thereby increasing the chance of hitting the correct hypothesis. We explore this
 256 hypothesized compensatory mechanism in Section 4.2.

257 4.2 Mechanistic explanations via modular analysis

258 We analyze the hypothesis generator of HPM-model to understand how individuals develop their
 259 feature strategies during learning. Fig. 5A, on the y-axis, the amount of two strategies—stochastic
 260 exploration (in dark purple) and posterior-based exploitation (in light purple)—generated as learning
 261 progresses (x-axis). Results of all participants are in Appendix B. This visualization reveals three
 262 distinct phases: Early phase, dominated by exploratory hypotheses, captures the initial random search

in the feature space; Middle phase, exhibiting fluctuations between explorative vs. exploitative dominance, suggesting an adaptive arbitration between sampling novel vs. validating evidence-driven candidates; Late phase is increasingly governed by the posterior-based exploitation, indicating a convergence toward the correct rule via gradient-like refinement. Interestingly, we found a significant negative correlation between the length of the middle phase and participants' memory capacity, as indexed by the fitted memory maintenance (w_0 parameter in memory module, see Section 3.2) of HPM-model (Fig. 5B). As shown, participants with lower memory capacity tend to sustain the exploration-exploitation alternation for longer periods. This finding demonstrates a rational-bonded hypothesis selection: individuals with more limited memory engage in prolonged cycles of exploration and exploitation to compensate for information loss and ultimately converge on the correct feature hypothesis.

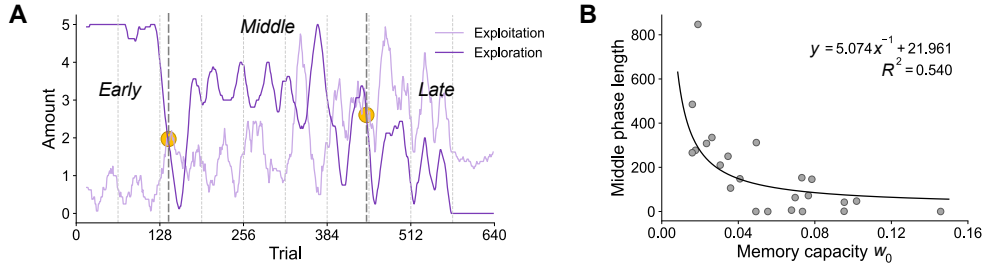


Figure 5: Dynamics of hypothesis selection strategies and their relation to memory capacity. (A) Temporal profiles of two strategies—stochastic exploration (dark purple) and posterior-based exploitation (light purple)—used by the HPM model across learning trials. (B) A significant negative correlation between the duration of the arbitration phase and individual memory capacity (w_0 from the memory module).

5 Discussion

Our study reveals the diverse category learning trajectories humans exhibit in a multi-dimensional feature space, shaped by individualized information processing (Fig. 3). Beyond reproducing choices, we recover trial-wise internal strategies that evolve over time (Fig. 4), using a modular Bayesian framework (Fig. 2) that balances exploratory and exploitative feature strategies under cognitive constraints (Fig. 5). Our findings highlight the heterogeneity of human learning and the importance of modeling internal processes, not just behavior, to understand adaptive cognition.

The mathematical foundation of our model conceptualizes learning as a dynamic information optimizer, combining both global grid-based flexible sampling and local gradient-driven optimization. This dual mechanism manifests particular strength in perceptual decision-making, where our formulation permits a high degree of flexibility in the partitioning of the stimulus space. Such theoretical flexibility supports the construction of a hypothesis space encompassing infinite decision rules – ranging from linear boundaries to complex nonlinear configuration [9] – as well as varying level of computational complexity (from simple 1-bit to high-dimensional multi-bit, [3]), and naturally extends to diverse conceptual structure (e.g., hierarchical, tree-like representations). By parameterizing task-specific internal states in a feature-based hypothesis space, our approach can bridge internal model dynamics with neural correlates of decision-making [28], probabilistic inference [23], and adaptive learning shifts [18], thereby advancing cognitive algorithms reverse-engineering from brain activity.

Limitation. Verbal reports provided ground truth for participants' internal representations but may have interfered with natural task behavior. However, these reports were used solely for model validation, not training. Thus, our model can reconstruct latent internal states even in tasks without explicit verbalization. Nonetheless, its generalizability should be tested across diverse task settings in future research.

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